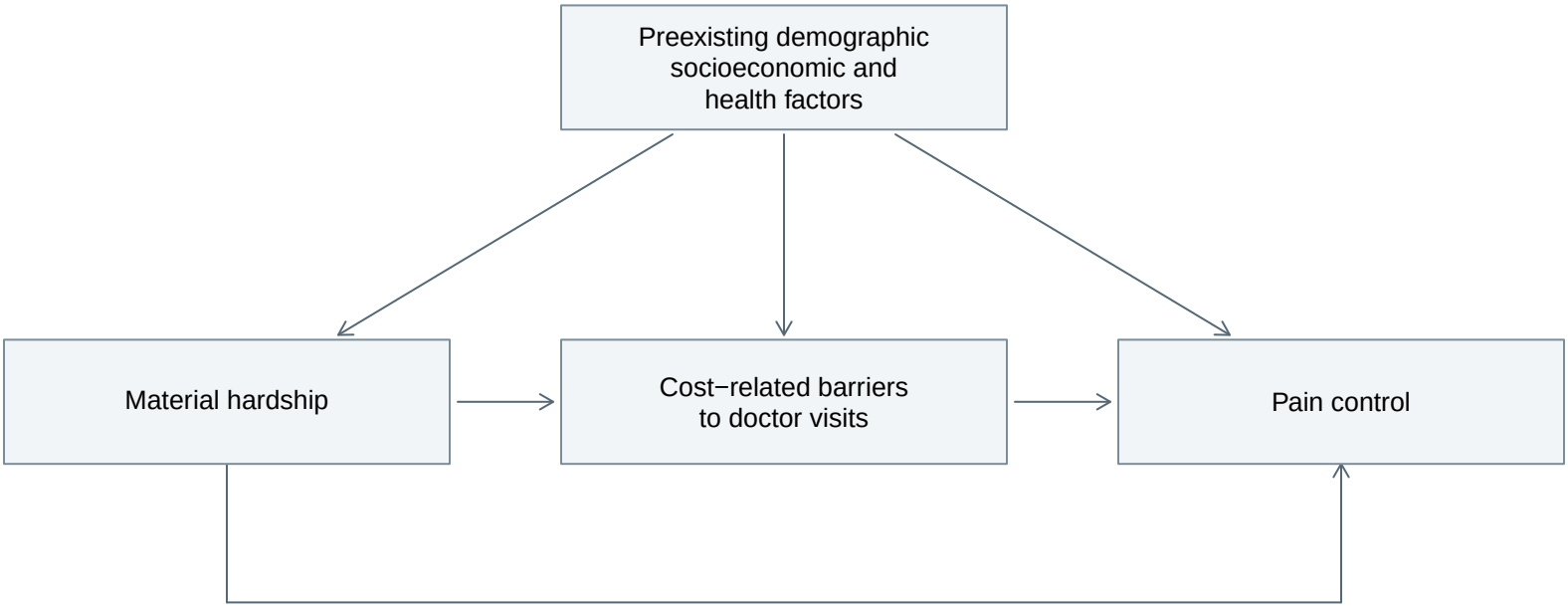Supplementary Figure S1. Hypothesized forward pathways